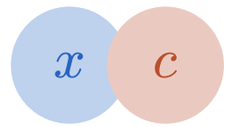
DFT's missing energy: same density, different electron-pair probabilities.

A **functional** maps the whole density to an energy. Kohn–Sham uses **noninteracting electrons at the same density**.





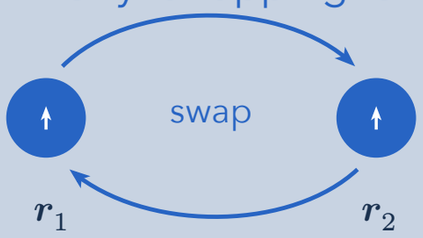


$$E[n] = \underbrace{T_s}_{\text{KS kinetic energy}} + \underbrace{V_{\text{ext}}}_{\text{nuclear attraction}} + \underbrace{E_H}_{\text{smooth-charge repulsion}} + \underbrace{E_{\text{xc}}}_{\text{the missing correction}}$$

$$E_{\text{xc}} = \underbrace{T - T_s}_{\text{kinetic correction}} + \underbrace{V_{ee} - E_H}_{\text{interaction correction}} = E_x + E_c$$

EXCHANGE

Antisymmetry: swapping fermions changes the sign.



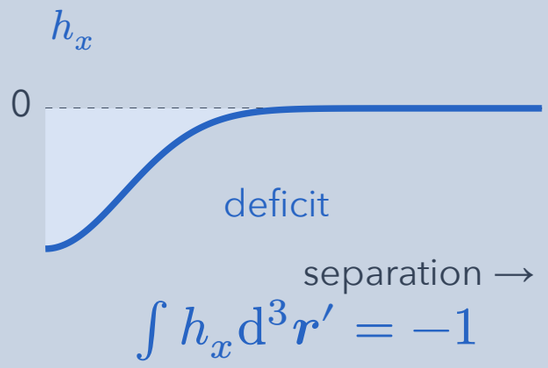
$$\Psi \to -\Psi$$

same position $\to \Psi = 0$

Pauli exclusion • same spin only • one determinant
Opposite-spin pairs have no exchange hole.



same-spin probability



$$\int h_x \mathrm{d}^3\boldsymbol{r}' = -1$$

One missing electron. No extra force.

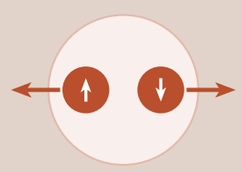
The hole persists without Coulomb repulsion.

$$E_x = \langle \Phi_s \mid \hat{V}_{ee} \mid \Phi_s \rangle - E_H \leq 0$$

Φ_s: antisymmetric KS orbital state. T, V_{ee}: exact interacting values at the same density.

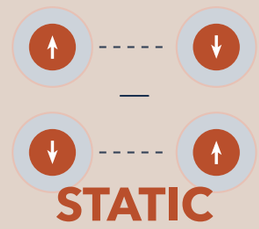
CORRELATION

Joint probabilities beyond the KS determinant.



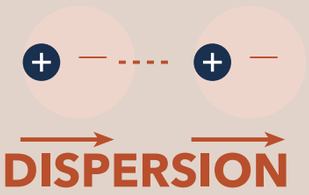
DYNAMIC

short-range avoidance and screening



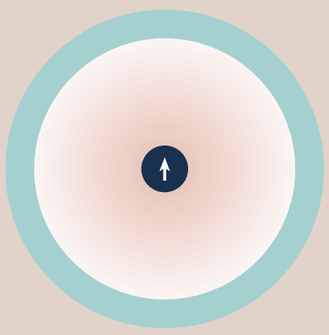
STATIC

stretched-bond singlet: several configurations

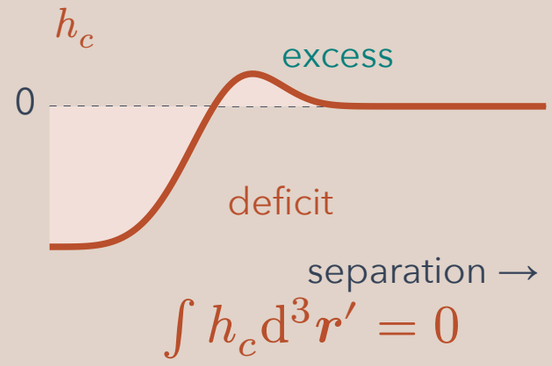


DISPERSION

coupled fluctuations, even at zero temperature



change in probability



$$\int h_c \mathrm{d}^3\boldsymbol{r}' = 0$$

Same-spin and opposite-spin pairs.

A near deficit is balanced by an excess farther away.

$$E_c = \underbrace{T - T_s}_{T_c \geq 0} + \underbrace{V_{ee} - E_H - E_x}_{U_c} \leq 0$$

Correlation balances reduced repulsion against a kinetic cost.

ONE REFERENCE ELECTRON

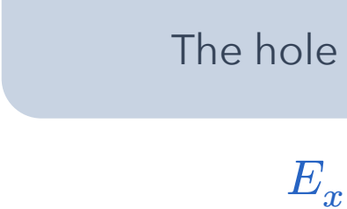
Only $N - 1$ others remain.

$$n_{\text{cond}} = n + h_x + h_c$$

net hole: $-1 + 0 = -1$

probability deficit, not an empty cavity

CLASSICAL CHARGES



correlated positions

Correlation: **yes**
Pauli exchange: **no**

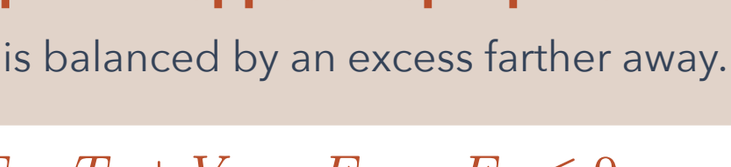
ONE ELECTRON NO SELF-REPULSION



$$E_H + E_x = 0$$
$$E_c = 0$$

Approximate XC can violate this cancellation.

XC FEEDS BACK INTO THE DENSITY



Why approximate? Exact XC is formally defined, but no practical general expression is known.

LDA: local electron gas · GGA: density gradients · hybrids: orbital exchange. Ordinary LDA/GGA miss long-range dispersion.

Schematic maps / curves; map center = reference electron. Hole sums are 3D integrals. Fixed nuclei; spin labels suppressed; atomic units; nuclear repulsion added separately.

Burke: The ABC of DFT, §§7, 11–13 · The exchange-correlation hole · Archer: Classical fluids, ch. 2